

Courier Management System

(COURIERTRACK) - Project Report

Domain: Courier & Logistics Management

Platform: Web Application / Database System

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Project Report

1. Executive Summary

The COURIERTRACK application is a purpose-built Courier Management System designed to centralize customer, courier, delivery, and tracking information. It maintains courier booking details, sender and receiver information, delivery status, and shipment records in one system. The application provides an organized way to register courier items, update delivery progress, and retrieve shipment information quickly.

2. System Architecture & Data Model

Object / Module	Type	Description
Customer	Parent / Master	Stores sender and receiver contact details and address information.
Courier / Shipment	Transactional Record	Stores courier booking, parcel details, source, destination, and tracking information.
Delivery / Tracking	Child / Activity	Records delivery status and updates associated with a courier shipment.

3. Activity Breakdown & Implementation Log

Module 1: System Environment Setup

Activity 1 & 2: Application Setup and Configuration

- Prepared the project environment and configured the database/application settings.
- Verified that the system modules and user access are ready for use.
- Configured the basic structure required for courier registration and tracking.

Module 2: Core Modules Creation

Activity 1: Customer Module

Created the customer module to maintain sender and receiver information, including name, phone number, email, and address.

Activity 2: Courier / Shipment Module

Created the shipment module for registering courier items and maintaining parcel information such as tracking number, package type, source, destination, and booking date.

- Created navigation options for Customer Registration.
- Created navigation options for Courier Booking.
- Created navigation options for Tracking and Delivery Status.

Module 3: Navigation & Application Configuration

Activity 1: Navigation Pages

Activity 2: Courier Management Application

Configured the application with dedicated navigation items so users can directly access customer, courier, and tracking functions.

- Customer ID
- Customer Name
- Phone Number
- Email ID
- Address
- Tracking Number
- Sender Details
- Receiver Details
- Parcel Type
- Weight
- Source and Destination
- Booking Date

Module 4: Data Fields & Relationships

Activity 1: Customer Profile Attributes

Activity 2: Courier Details & Customer Link

- Total Couriers: COUNT of registered courier shipments.
- Delivered Couriers: COUNT of shipments marked as Delivered.
- Pending Couriers: COUNT of shipments that are still in transit or pending.

Module 5: Automated Calculations & Status Management

Activity 1: Courier Summary Information

Activity 2: Delivery Status

Implemented shipment status values such as Booked, In Transit, Out for Delivery, Delivered, and Cancelled to make the current courier condition easy to identify.

Module 6: Automation & Tracking Deployment

Activity 1: Tracking Interface

Constructed an interactive tracking interface where users can enter a tracking number and retrieve the related courier information and latest delivery status.

Activity 2: Delivery Update

Provided an organized method for authorized users or administrators to update shipment status and maintain delivery records.

- Register sender and receiver information.
- Create a courier shipment record.
- Generate or assign a unique tracking number.
- Store parcel and destination details.
- Set the initial shipment status as Booked.
- Search using the tracking number.
- Display sender and receiver details.
- Display source and destination.
- Display current delivery status.
- Maintain updated tracking information.

Module 7: Courier Booking Process

Module 8: Shipment Tracking

- View registered customers and courier shipments.
- Update courier delivery status.
- Maintain accurate shipment records.
- Search and retrieve courier information when required.
- Reduce manual record-keeping work.
- Validate required customer and courier fields.
- Prevent incomplete shipment registration.
- Provide appropriate access to management functions.
- Verify that tracking information is displayed correctly.

Module 9: Administration & Record Management

Module 10: Validation & User Access

- Customer registration was checked.
- Courier booking was checked.
- Tracking number search was checked.
- Delivery status updates were checked.

Module 11: Testing & Verification

Activity 1: Functional Verification

Activity 2: Data Verification

The system records were reviewed to confirm that customer, shipment, and tracking information are stored and displayed consistently.

4. Conclusion & Verification

The Courier Management System provides a structured solution for managing courier bookings, customer details, shipment information, and delivery tracking. The modules and activities are designed to support accurate record management, easier tracking, and efficient courier operations.

- Centralized courier and customer information.
- Simplified courier booking and record maintenance.
- Unique tracking-based shipment identification.
- Easy delivery status monitoring.
- Reduced dependence on manual record keeping.
- Improved accessibility of courier information.

5. Project Outcome

6. Final Verification

All major modules of the Courier Management System were reviewed for data entry, record management, shipment tracking, and delivery-status handling. The project structure follows the same report-style organization as the supplied reference document, adapted to the Courier Management domain.